CSE232_CN_Mid-Sem Exam (20 Marks)

1. Alice wants confidentiality, so Alice encrypts an e-mail using a randomly generated symmetric key K_S, then encrypts K_S using Bob's public key before transmitting the e-mail. (2 marks)
   a. Compare this hybrid approach with encrypting the entire message using Bob's public key. (1 mark)
   b. Suppose Alice uses only Bob's public key to encrypt a message. For each property, state whether confidentiality, integrity, and authentication are achieved, and justify the answer. (1/3 * 3 = 1 mark)

2. Reliable Data Transfer introduces checksums, acknowledgements (ACK and NAK), sequence numbers, retransmissions, and timers. Consider an unreliable channel that can corrupt data packets or acknowledgements, lose data packets or acknowledgements, and delay acknowledgements long enough to cause premature timeouts. (5 marks)
   a. Analyze why each mechanism is necessary, identify the failure it addresses, and explain how the mechanisms collectively provide reliable delivery. (3 marks)
   b. Distinguish between a lost packet and a delayed packet from the sender's perspective. Explain how duplicate packets can arise and how the receiver prevents duplicate delivery to the application. (2 marks)

3. Alice wants confidentiality, integrity, and authentication simultaneously. Describe the complete sequence of operations at Alice's side and Bob's side in the correct order and using correct notation. Explain how each security requirement is met. (5 marks)

   Assume access to public-key cryptography, symmetric-key cryptography, and hash functions. Use standard notation such as $H(m)$, K_s^{AB}, K^-_A, K^+_A, etc., or clearly define equivalent notation. No partial marks will be given if all three conditions are not achieved.

4. An rdt3.0 stop-and-wait sender operates over a 1 Gbps link with a 15 ms one-way propagation delay and 8,000-bit packets. Assume negligible ACK transmission time and no packet loss during measurement. (5 marks)
   a. Calculate the packet transmission delay, RTT, sender utilization, and effective throughput in bits/s. (2 marks)
   b. Determine the minimum pipeline window N needed to keep the sender continuously busy, assuming N packets may be sent without waiting for their acknowledgements. (3 marks)

5. Mark True or False. Correct answer will get 0.2 marks. Incorrect answer will penalize with -1 mark.
   a. Root DNS servers store the IP address of every host on the Internet.
   b. During iterative DNS resolution, the local DNS server may contact multiple DNS servers before obtaining the final IP address.
   c. If a local DNS server already has a valid cached mapping for a hostname, the local DNS server will still confirm it by contacting root, TLD, or authoritative servers before responding.
   d. In iterative DNS resolution, a DNS server may reply with the address of another DNS server rather than the final IP address requested.
   e. DNS caching scheme always guarantees that users receive the most recent IP address of a host.

6. Why is SMTP considered a push protocol, and how does this differ from HTTP's pull-based communication model? (1 + 1 marks)

Quiz 1 (10 Marks), CSE232-SecA

1. Compare Non-persistent HTTP and Persistent HTTP (without pipelining). Which one is more efficient for loading a webpage with 10 embedded images, and why?

2. A 100 KB file is sent over a 1 Mbps link. Calculate the transmission delay in milliseconds. (Assume 1 KB = 1000 bytes)

3. How many RTTs (Round Trip Times) are required in total to retrieve an HTML file and 4 images using Non-persistent HTTP? (Exclude the initial DNS lookup and transmission times).

4. In Persistent HTTP (without pipelining), after the initial connection is established and the base HTML file is received, how many additional RTTs are needed to fetch 3 embedded objects?

5. A network has a cache hit rate of 40%. The access time to the local cache is 10ms, while fetching from the origin server takes 100ms. Calculate the average response time.

6. Assume a link has a propagation speed of 2×10^8 m/s and a length of 2,000 km. If a packet is 1,000 bits and the bandwidth is 1 Mbps, find the ratio of propagation delay to transmission delay.

7. A campus access link has a capacity of 10 Mbps. The average request size is 100,000 bits and the internal request rate is 120 requests/second. Calculate the minimum cache hit rate required to prevent the access link from being overloaded (i.e., to keep traffic below 10 Mbps).

8. A company installs a web cache to save money on bandwidth. If their current WAN usage costs $1000/month and the cache achieves a 60% hit rate, what is the new estimated monthly cost?

9. Consider a scenario where a popular news video (100 MB) is requested by 50 different students on a campus. Contrast the total bandwidth consumed on the university's external internet link with and without a web proxy.

10. A client uses Persistent HTTP (no pipelining) to fetch a 10 KB HTML file and two 50 KB images. Link rate is 1 Mbps and RTT is 100ms. Calculate the total time **from the first request** until all data is received. (Ignore handshake RTT and assume 1 KB = 8000 bits).

5/10/26

Selective Repeat (SR)

- Packet lost झाल्यास फक्त त्याच packet ची retransmission होते, not the whole sequence.

- receive pkt 3, buffer, send ack 3

If time out / ACK received ⇒ sender
in order (out of order ⇒ Receiver
→ put in buffer & send.

Dilemma:

0, 1, 2, 3 → 4 nos
window size = 3

sender: 0 → 1 → 2 | receive: ACK
3 → 0 → 1
receiver can distinguish

dilemma:

Say packets 0, 2 were sent, but Ack / packets get lost in such a way that sender's window moves around. Eventually sender will send 0 again.

So window & sequence size are dependent! (SO?)

for N=3, 0 1 2 3 4 5 6 (at least double size), we won't face this problem

Everything reaches in order ✓ Not security wise ✗.

Selective Repeat → TCP

→ Point-to-Point (one sender ↔ one receiver)

Client TCP Server

→ Reliable, in order byte stream
A B C D E F G H --- (conts byte stream)

→ Full duplex : both sides can send data simultaneously (same time)

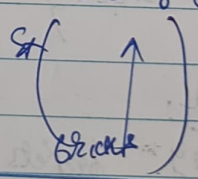
Simplex / duplex / full duplex

→ MSS

→ Cumulative ACKs ✓

→ Pipelining : many segments in flight.

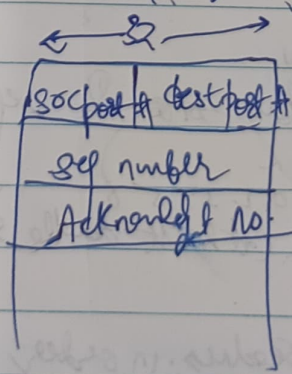
→ Flow Control : don't send faster than the rcvr can handle.
congestion Control.

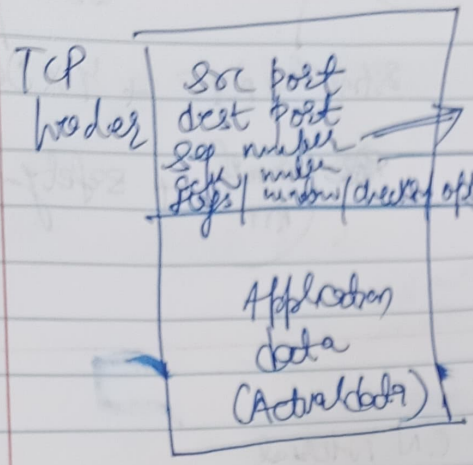


→ UDP: src port, dest port, len, chksum

→ TCP header :

(TCP is full duplex)





byte number of first

ACK=500 ⇒ 0–499 ✓

Navigated by the OS.

checksum

GET / index.html.

(TCP puts in data/payload portion)

Segment 1

SSQ = 100

Data = 500B ⇒ 100 ---- 599

Next seq / SSQ = 600. (next that I expect from you).

Left:	Middle	Right
already ACKed	Acceptable bytes. (Size = N)	Not usable yet.

As ACKs arrive, window slides fwd.

TCP RTT ⇒ sending the segment & receiving ACK

Simple RTT: → just measured

estimated RTT → in past measured (smoothed average)

⇒ (1−α) Estimated RTT + α · SimpleRTT.

Blue → actual

Pink → for certain alpha

(best estimate of normal RTT)

(how much is actual RTT fluctuating and deviating from RTT)

Timeout interval = estimated RTT + 4 × DevRTT

(Normal RTT + safety margin)

CN Tutorial

13/10/26

Q.1: TCP congestion window (cwnd) for TCP Tahoe and TCP Reno.
Triple duplicate ACK: Round 8
Timeout: Round 16.
X: Transmission round.
Y: cwnd. in MSS (Maximum Segment size).
units.

What is TCP congestion control?
Road capacity limited → send. tracks slowly.
If packet loss

(1) Congestion window.
(2) Max Seg Size
(3) Slow start Threshold. (ssthresh)
cwnd < ssthresh → Slowstart
cwnd ≥ ssthresh → Fast

(B2) ssthresh → Tahoe, Reno.
cwnd → Tahoe. → Reno.

$$UR = \left[\frac{N/R}{1/R + 1/R_2} \right]$$

Graph 06:
R1–5 : window grow.
R6–8 : window slowly grow.
R8 : Tahoe & Reno diff behaviour.
R16 : Both reset (after timeout)

Loss → Time out.
→ Triple Duplicate ACK
congestion avoidance.

slow start & congestion avoidance.
cwnd : exponential grow.
linear grow.

(B1)
(B2) ssthresh → Tahoe.
→ Reno.

cwnd
Tahoe
Reno.

Before loss
cwnd = 20 MSS
ssthresh = 20/2 = 10 MSS (for both).

cwnd after loss recovery,
Tahoe : 1 MSS
Reno : 10 MSS.

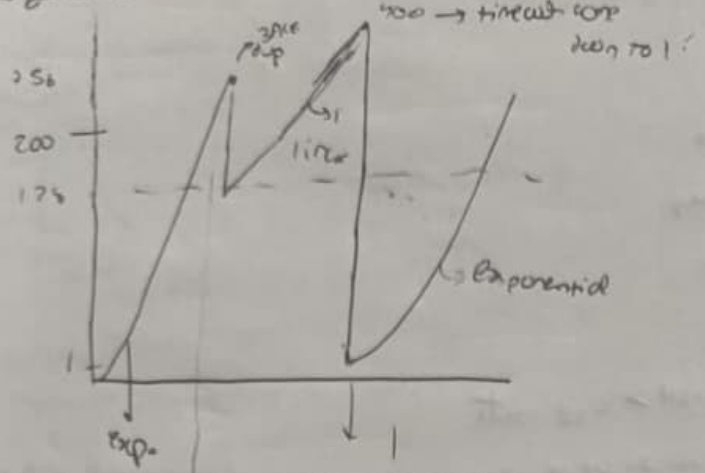
(B3) Tahoe: slow start → window grows exponentially
Reno: congestion avoidance → window grows linearly

(B4) After R8 loss, Tahoe starts with cwnd = 1 and ssthresh = 10 MSS
: window increases during slow start. until threshold. & switches to congestion avoidance. (Round 8–14).

(B5) Timeout in R16. Tahoe ≈ 13 MSS.
and
Reno cwnd ≈ 18 MSS.

→ Reno :- if loss → 3DupACK → drop down by half otherwise ↓

CWND → Congestion window size.



C_1	C_2
1	1
2	2
4	4
8	8
16	16
32	32
64	64 → loss

Say C_1 → loss prior to C_2

→ 32 128 → C_2 has higher probability of loss since it is sending more packets.

16 256 → if C_1 → loss.

but now more chance of C_2 having a loss

So, 16 128

Congestion control
↳ Asynchronous → sender doesn't communicate with anyone to reduce congestion
↳ Fair

$\frac{w}{2} \rightarrow \frac{w}{2} + 1$

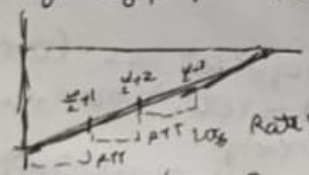
$w = 5$ means 5 MSS are being sent

MSS → max. segment size

↳ Amount of data sent in a RTT

↳ Asynchronous & fair

→ Average Throughput per RTT in TCP :- → sigmoid + exponential term similar to small throughput



average throughput = $\frac{\frac{w}{2} + w}{2} = \frac{3w}{4}$ per RTT

Rate = $\frac{\#\text{lost packet}}{\#\text{Total packet}}$

packets → $\frac{w}{2} + \left(\frac{w}{2}+1\right) + \left(\frac{w}{2}+2\right) \cdots \frac{w}{2} + \frac{w}{2}$

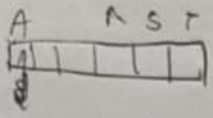
→ $\left(\frac{w}{2}+1\right)\frac{w}{2} + \frac{\frac{w}{2}(w+1)}{2}$

$L = \frac{1}{3w^2}$ ← now loss rate = $\frac{1}{\frac{3w^2}{8}}$

$\frac{3w^2}{8} + \left(\frac{3w^2}{4}\right)$ ← right

→ w → no. of segments → Cubic

→ In 2 way handshake the client might retransmit conn. req. before it receives the ack. So 3 way handshake is needed.



1 → first way

1 → second way

1 → third way

+ Congestion control:-

" happens when arrival rate > service rate

→ if you send 10 packets out of which one is retransmission → actual throughput is 9 packets.

= End to end congestion control → router doesn't tell source there is congestion

TCP congestion control

↳ 1 → 2 → 4 → 8 . . . 32 → 1 → 2 → . . . 5 → 1 → 2 . . .

↓ say congestion here

→ congestion (packet dropped)

↓ go back to

human congestion control

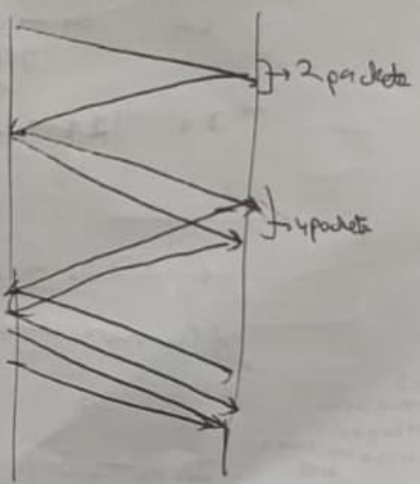
congestion

1 → 2 → 4 . . . 32 → 16 → 32 → 16 → 8 → 4

↓ go to half

congestion

MSS → how many bytes it can send.

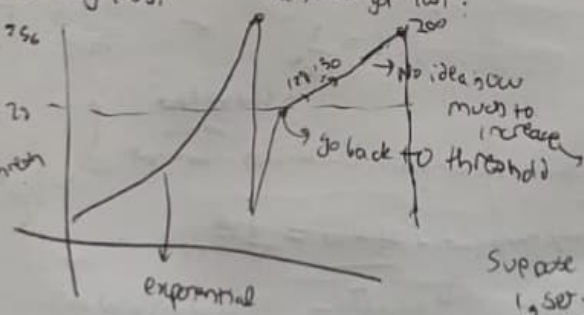


TCP detects loss via

↳ Timeout → the worst scenario where packet went

↳ 3 duplicate ACK ↳ this is better since only one packet got lost.

TAHO → 1st cut tol. w any loss.



Suppose at 200, we get loss → come to 1, set threshold to 100. So from 1, exponential till 100 then linearly.

Suppose after that we go to 700 & then congestion happens → go to 1, set threshold to 200.

TCP

Flow control

TCP talks with sender/self accordingly sends data

Receive window field is used

Receive window can be set during handshake

receive window

Periodic, a special msg, receive window will never be 0 it should be min 1 even if it is 0 send 1

So if it is 0 sender will send one byte of data

Then it is to be the case while it will atleast one byte of buffer will be empty.

TCP connection management

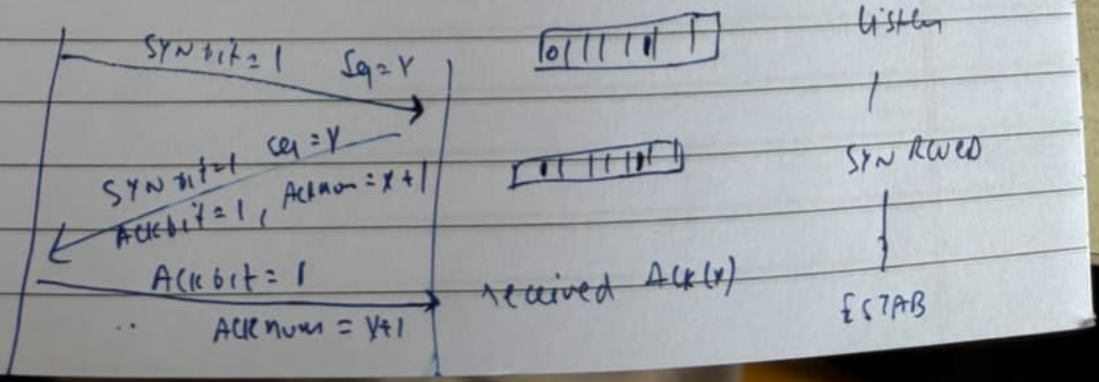
we st with a seq no. or ack no.

~~if we st will a seq no. 0, 1 then~~

we start with a random seq no & it can be communicated in SYN ACK SYN ——.

In 2 handshake

The server does not know whether sender received it msg or not



Closing a TCP connection

FIN bit = 1

•

Principles of Congestion control:

Congestion control
if arrival rate >> rate of service



Our output becomes 0

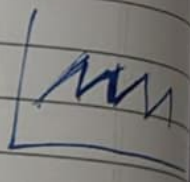
End - end congestion control:

ATM ← take help from routers

Slow start
Congestion control
Fast recovery

} def want to be asked

Loss detection :-



3 duplicate Ack :

- ssthresh is reset to 1/2 of the cwnd
- TCP Tahoe sets cwnd to 1
TCP Reno sets cwnd to ssthresh + 3MSS
(1 of 3 dup acks)

fundamental Problem with TCP:
If ~~any~~ a packet loss happens TCP through will never reach its full potential again.
TCP also does not know why packet loss happens bcz if it was not bcz of congestion still TCP will reduce the cwnd but it was not required.

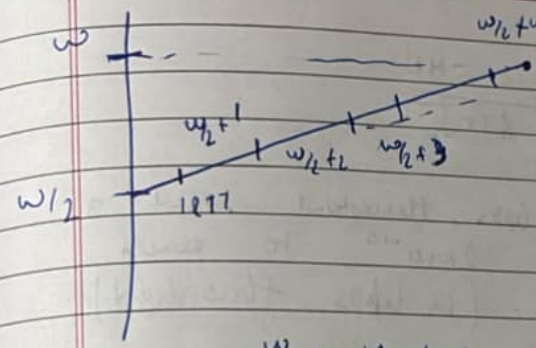
Avg throughput: (We only take the linear growth period, tail of exp hand)

$$\min = \frac{1}{2}\frac{W}{RTT} \qquad \max = \frac{W}{RTT}$$

$$\text{Avg} = \frac{1}{RTT}\left(\frac{\frac{W}{2} + W}{2}\right) = \frac{3}{4}\frac{W}{RTT}$$

$\frac{W}{2}$ / RTT

R



$$\frac{w}{2} + \frac{w}{2}+1 + \cdots \cdots \frac{w}{2}+\frac{w}{2} \Rightarrow \left[\frac{w}{2}\right]$$

$$\left(\frac{w}{2}\right)\left(\frac{w}{2}+1\right) + \frac{w/2\left(\frac{w}{2}+1\right)}{2}$$

$$\frac{2w^2}{8} + \frac{2w}{4} + \frac{w^2}{8} + \frac{w}{4}$$

$$\frac{3w^2}{8} + \frac{3w}{4} \quad \text{— small so ignore}$$

$$\approx \frac{3w^2}{8}$$

So over the period we sent $\approx 3w^2/8$ packet

(So we dt only seeing time this & we assum only one source to dest & packet loss will happen when it reaches max throughput

$$L = \frac{1}{\frac{3w^2}{8}}$$

$$T = \frac{3}{4}\frac{w}{RTT}$$

TCP throughput : $\frac{1.22 \cdot MSS}{RTT\sqrt{L}}$

to achieve a 10 Gbps throughput, need a loss rate of $L = 2 \times 10^{-10}$ to reach its max potential. (10 Gbps throughput)

So it is like if the first loss happens early then again the threshold will be small so it will be linear increase for a large period of time.

- Software defined network did not work in Internet but is used in data centers & worked well

Cubic

So it inc.'s the w with distance of cwnd from w & if dist is more it acc cwnd more ---

TCP Vegas :

Sender measures RTT... if less

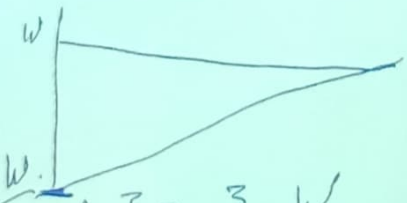
IF RTT ~~is~~ dec. or close to what is best possible so which means network ok..

If RTT is inc.. then it means there is queue is being formed at some routers so dec its sender dec its rate

TCP conn can also be left half open.

No ~~no~~ time guarantee

Avg. throughput



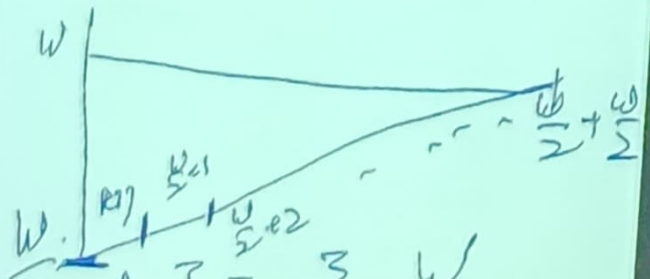
Avg $T_h = \frac{3}{4} \frac{W}{RTT}$

Untitled Page
No additional text

TCP Throughput
You're right. OneNote e...

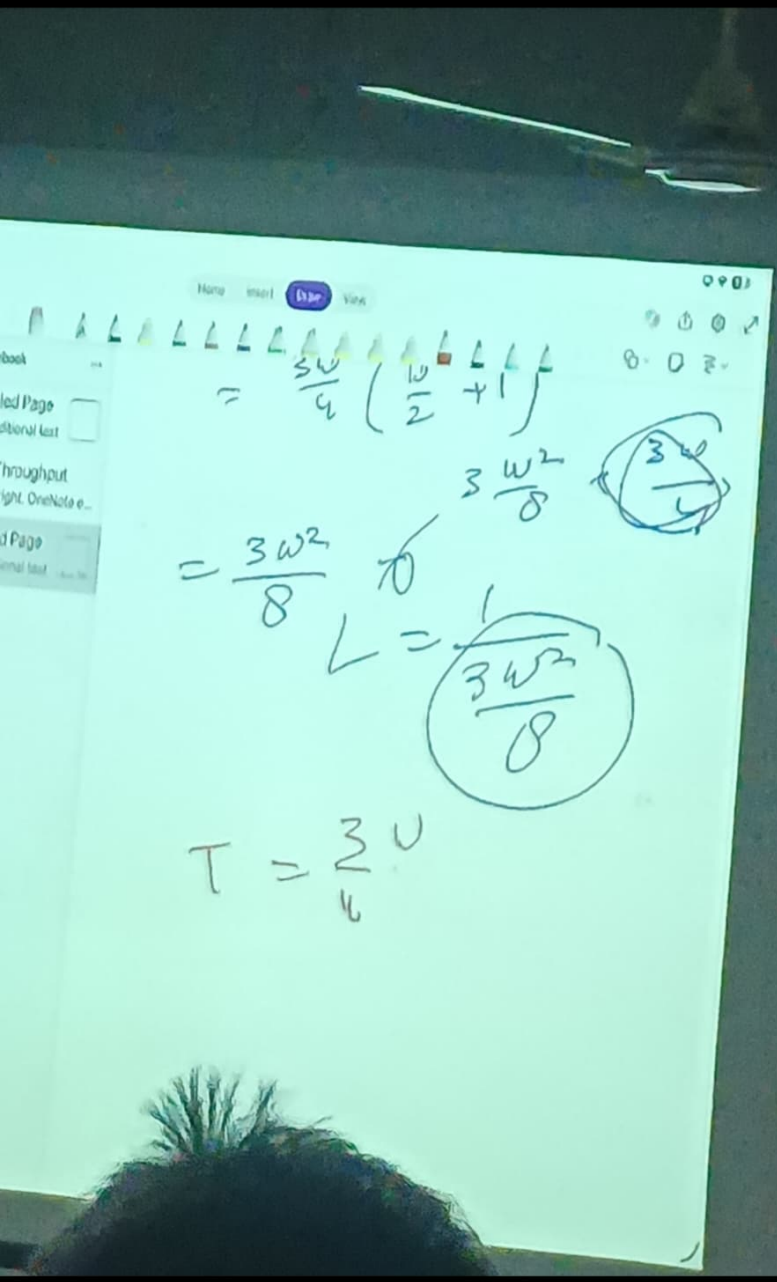
Untitled Page
No additional text

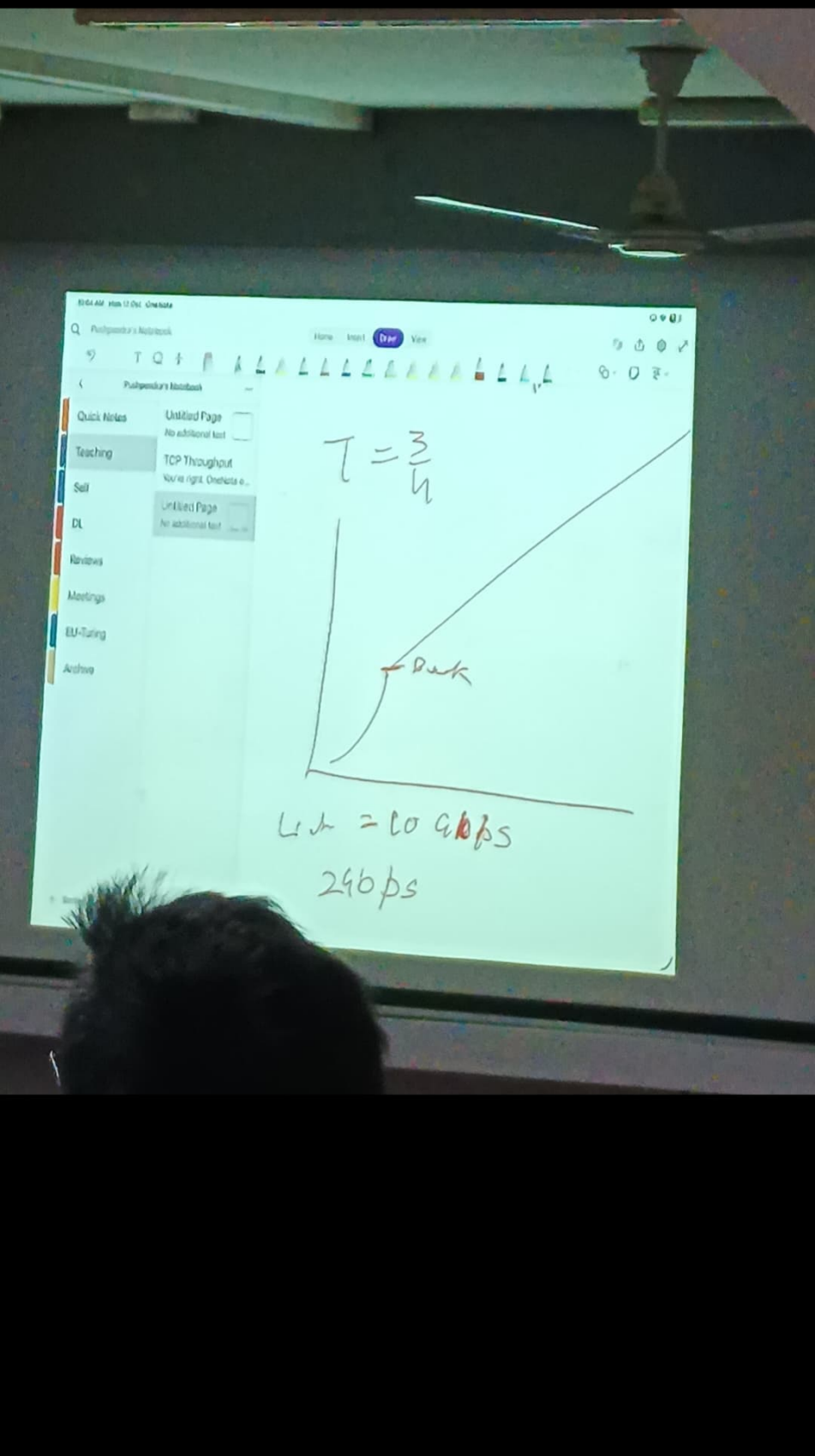
Avg. throughput



Avg $T_h = \frac{3}{4} \frac{W}{RTT}$

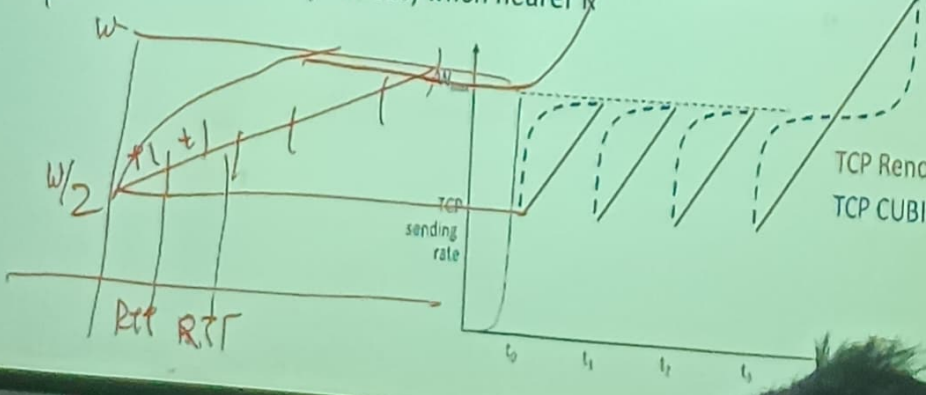
Loss Rate $= \frac{2}{W}$


Home
Insert
Draw
View
= 3w/4 (w/2 + 1)
3w²/8
= 3w²/8 ≥ 0
L = 1
3w²/8
T = 3w/4


Quick Notes
Teaching
Skill
DL
Reviews
Meetings
ELI-Tuning
Archive
Untitled Page
No additional text
TCP Throughput
Untitled Page
No additional text
T = 3/4
Dark
240 pps

TCP CUBIC

- K: point in time when TCP window size will reach W_{max}
  - K itself is tunable
- increase W as a function of the *cube* of the distance between c time and K
  - larger increases when further away from K
  - smaller increases (cautious) when nearer K



8:30 ... 0.03 KB/s 5G 24
Vidush IIITD
19 September, 2:47 pm

b) The number of segments is $\left\lceil \frac{2^{32}}{536} \right\rceil = 8{,}012{,}999$. 66 bytes of header get added to each segment giving a total of 528,857,934 bytes of header. The total number of bytes transmitted is $2^{32} + 528{,}857{,}934 = 4.824 \times 10^{9}$ bytes. Thus it would take 249 seconds to transmit the file over a 155~Mbps link.

- Correct argument and correct calculation leads to 2 + 2. Correct calculation without argument can also be given 2 + 2. However, if the calculation is wrong then decide on the argument and you may assign either 0 or 1.

5. Let's consider the delay introduced by the TCP slow-start phase. Consider a client and a Web server directly connected by one link of rate *R*. Suppose the client wants to retrieve an object whose size is exactly equal to 15 *S*, where *S* is the maximum segment size (MSS). Denote the round-trip time between client and server as RTT (assumed to be constant). Draw the transmission showing message communication and ignoring protocol headers, determine the time to retrieve the object (including TCP connection establishment) when

   a. 4 S/R>S/R+RTT>2S/R (2 marks)
   b. S/R+RTT>4 S/R (2 marks)
   c. S/R>RTT (2 marks)

Answer: The transmission diagram will remain same for all three cases and if only

Reply

8:30 पे ···
0.03 KB/s VoLTE 5G 24
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8. Describe confidentiality, authentication, and message integrity, precisely and succinctly and how they are achieved. (1 x 3 = 3 marks)

Confidentiality: If A is sending to B a message m then no other should understand the message. It is achieved through Encryption.

Authenticity: A should be able to authenticate B, i.e., confirm the identity of B or that B is what B claims to be. It is achieved through digital signature.

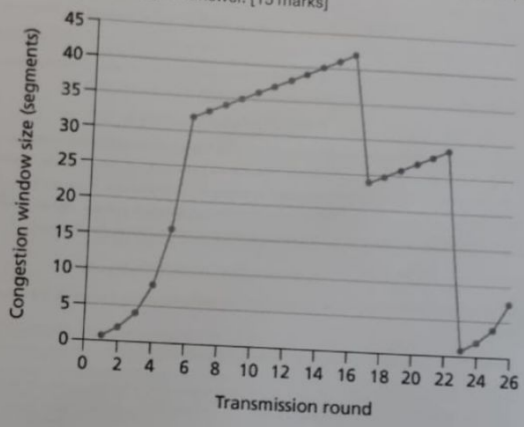
- Integrity: A message m should not be tampered or if it is tampered then the recipient should be able to figure out that message has been tampered. It is achieved through Hashing.

Reply ❤️ 🙏

8:30 ... 4.16 KB/s VoLTE 5G 24

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19 September, 2:47 pm

6. Consider the figure given. Assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions. In all cases, you should provide a short discussion justifying your answer. [15 marks]



a. Identify the intervals of time when TCP slow start is operating. [1 mark]

b. Identify the intervals of time when TCP congestion avoidance is operating. [1 mark]

c. After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout? [1 mark]

d. After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout? [1 mark]

e. What is the initial value of `ssthresh` at the first transmission round? [1 mark]

f. What is the value of `ssthresh` at the 18th transmission round and why? [1+1 mark]

Reply

8:30 ... 4.16 KB/s 5G 23
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i) The threshold will be set to half the current value of the congestion window (8) when the loss occurred and congestion window will be set to the new threshold value + 3 MSS. Thus the new values of the threshold and window will be 4 and 7 respectively.
j) threshold is 21, and congestion window size is 1.
k) round 17, 1 packet; round 18, 2 packets; round 19, 4 packets; round 20, 8 packets; round 21, 16 packets; round 22, 21 packets. So, the total number is 52.

- As per the marks, either full or zero in each point.

7. Consider sending a large file from a host to another over a TCP connection that has no loss. (2 + 2 marks)
   a. Suppose TCP uses AIMD for its congestion control without slow start. Assuming *cwnd* increases by 1 MSS every time a batch of ACKs is received and assuming approximately constant round-trip times, how long does it take for *cwnd* increase from 6 MSS to 12 MSS (assuming no loss events)?
   b. What is the average throughout (in terms of MSS and RTT) for this connection up through time=6 RTT?

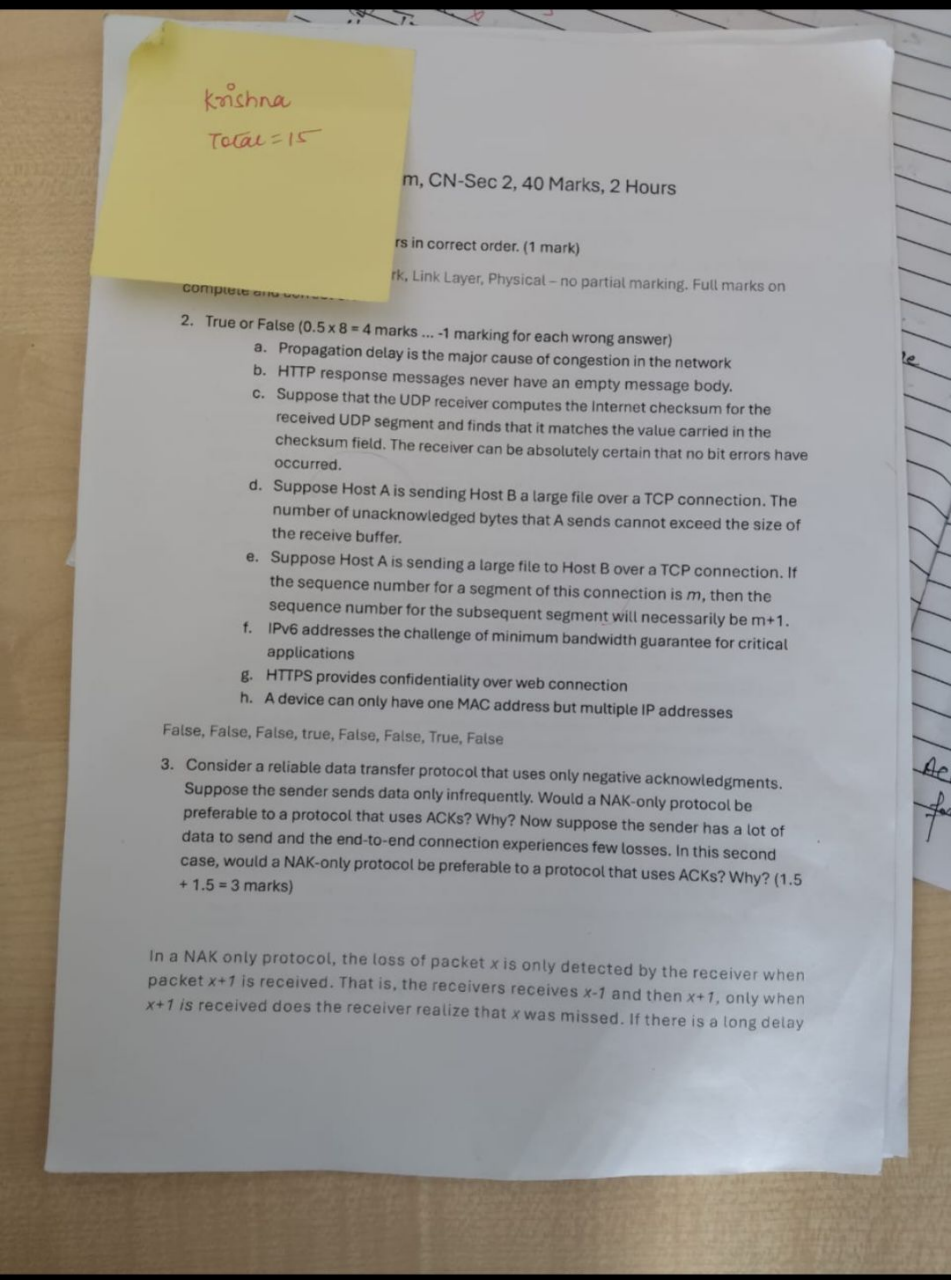
a) It takes 1 RTT to increase CongWin to 7 MSS; 2 RTTs to increase to 8 MSS; 3 RTTs to increase to 9 MSS; 4 RTTs to increase to 10 MSS; 5 RTTs to increase to 11 MSS; 6 RTTs to increase to 12 MSS.

b) In the first RTT 6 MSS was sent; in the second RTT 7 MSS was sent; in the third RTT 8 MSS was sent; in the fourth RTT 9 MSS was sent; in the fifth RTT, 10 MSS was sent; and in the sixth RTT, 11 MSS was sent. Thus, up to time 6 RTT, 6+7+8+9+10+11 = 51 MSS were sent. Thus, we can say that the average throughput up to time 6 RTT was (51 MSS)/(6 RTT) = 8.5 MSS/RTT.

- Correct reasoning but incorrect answer may be given 1 out of 2 marks else give full marks if both are correct.

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8:30 पे ···
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krishna
Total = 15

m, CN-Sec 2, 40 Marks, 2 Hours

rs in correct order. (1 mark)

rk, Link Layer, Physical – no partial marking. Full marks on complete and

2. True or False (0.5 x 8 = 4 marks ... -1 marking for each wrong answer)
   a. Propagation delay is the major cause of congestion in the network
   b. HTTP response messages never have an empty message body.
   c. Suppose that the UDP receiver computes the Internet checksum for the received UDP segment and finds that it matches the value carried in the checksum field. The receiver can be absolutely certain that no bit errors have occurred.
   d. Suppose Host A is sending Host B a large file over a TCP connection. The number of unacknowledged bytes that A sends cannot exceed the size of the receive buffer.
   e. Suppose Host A is sending a large file to Host B over a TCP connection. If the sequence number for a segment of this connection is *m*, then the sequence number for the subsequent segment will necessarily be m+1.
   f. IPv6 addresses the challenge of minimum bandwidth guarantee for critical applications
   g. HTTPS provides confidentiality over web connection
   h. A device can only have one MAC address but multiple IP addresses

False, False, False, true, False, False, True, False

3. Consider a reliable data transfer protocol that uses only negative acknowledgments. Suppose the sender sends data only infrequently. Would a NAK-only protocol be preferable to a protocol that uses ACKs? Why? Now suppose the sender has a lot of data to send and the end-to-end connection experiences few losses. In this second case, would a NAK-only protocol be preferable to a protocol that uses ACKs? Why? (1.5 + 1.5 = 3 marks)

In a NAK only protocol, the loss of packet *x* is only detected by the receiver when packet *x+1* is received. That is, the receivers receives *x-1* and then *x+1*, only when *x+1 is* received does the receiver realize that *x* was missed. If there is a long delay

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Vidush IIITD
19 September, 2:47 pm

g. What is the value of `ssthresh` at the 24th transmission round and why? [1+1 mark]

h. During what transmission round is the 70th segment sent and why? [1 +1 mark]

i. Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of `ssthresh` and why? [1 +1mark]

j. Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the `ssthresh` and the congestion window size at the 19th round? [1 mark]

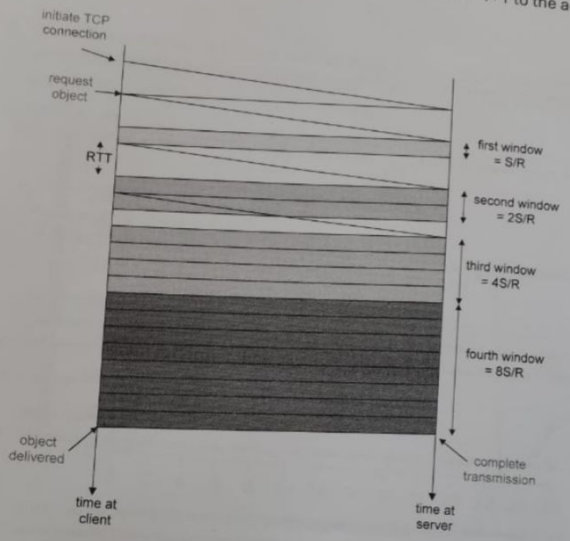
k. Again suppose TCP Tahoe is used, and there is a timeout event at 22nd round. How many packets have been sent out from 17th round till 22nd round, inclusive? [1 mark]

a) TCP slowstart is operating in the intervals [1,6] and [23,26]
b) TCP congestion avoidance is operating in the intervals [6,16] and [17,22]
c) After the 16th transmission round, packet loss is recognized by a triple duplicate ACK. If there was a timeout, the congestion window size would have dropped to 1.
d) After the 22nd transmission round, segment loss is detected due to timeout, and hence the congestion window size is set to 1.
e) The threshold is initially 32, since it is at this window size that slow start stops and congestion avoidance begins.
f) The threshold is set to half the value of the congestion window when packet loss is detected. When loss is detected during transmission round 16, the congestion windows size is 42. Hence the threshold is 21 during the 18th transmission round.
g) The threshold is set to half the value of the congestion window when packet loss is detected. When loss is detected during transmission round 22, the congestion windows size is 29. Hence the threshold is 14 (taking lower floor of 14.5) during the 24th transmission round.
h) During the 1st transmission round, packet 1 is sent; packet 2-3 are sent in the 2nd transmission round; packets 4-7 are sent in the 3rd transmission round; packets 8-15 are sent in the 4th transmission round; packets 16-31 are sent in the 5th transmission round; packets 32-63 are sent in the 6th transmission round; packets 64 – 96 are sent in the 7th transmission round. Thus packet 70 is sent in the 7th transmission round.

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8:30 ... 3.96 KB/s 5G 23
Vidush IIITD
19 September, 2:47 pm

transmission diagram is correct then give 1+1+1 to it. Rest 1+1+1 to the actual answers.



a) Referring to the figure below, we see that the total delay is
RTT + RTT + S/R + RTT + S/R + RTT + 12S/R = 4RTT + 14 S/R

b) Similarly, the delay in this case is:

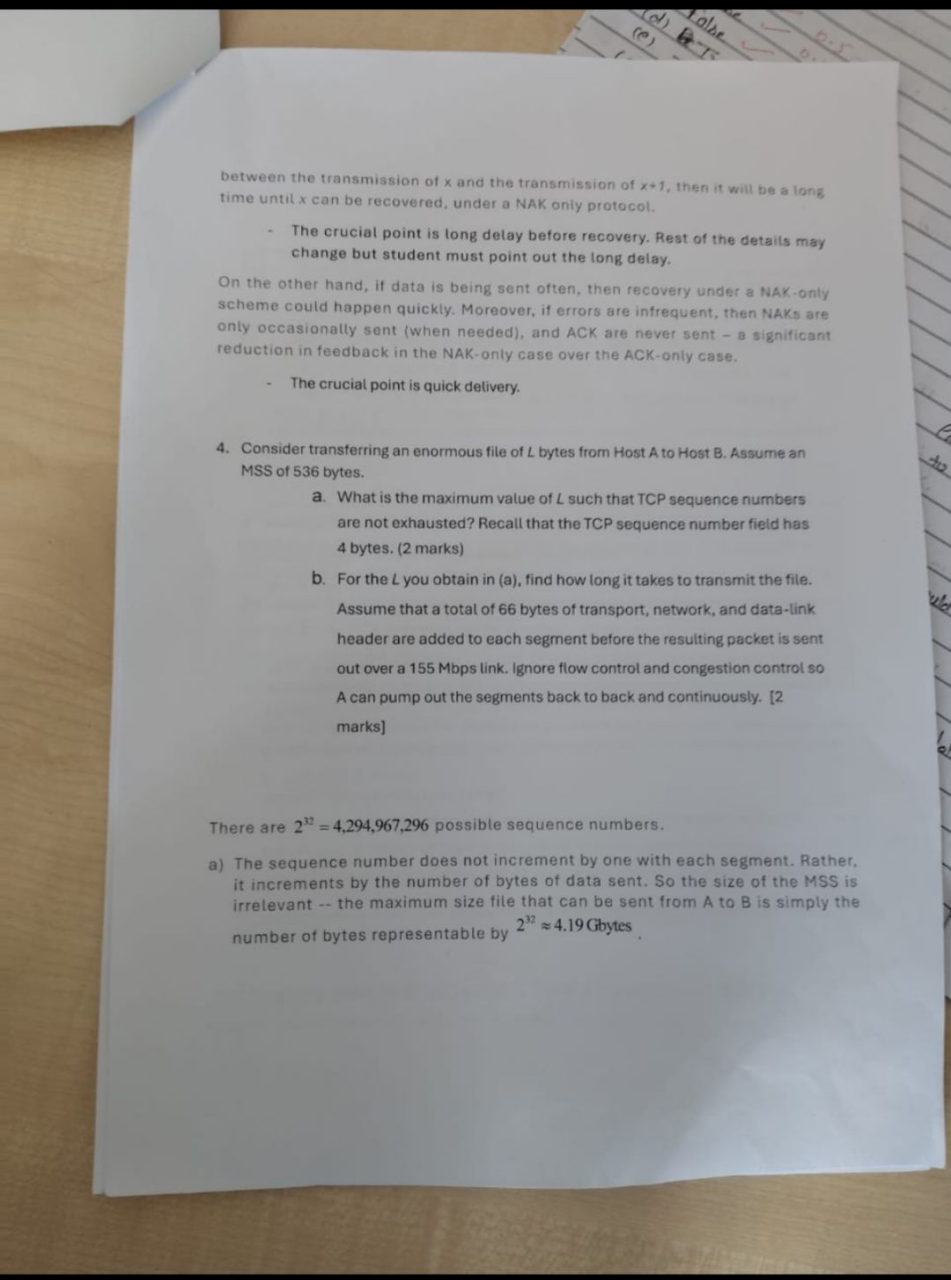
RTT+RTT + S/R + RTT + S/R + RTT + S/R + RTT + 8S/R = 5RTT +11 S/R

c) Similarly, the delay in this case is:

RTT + RTT + S/R + RTT + 14 S/R = 3 RTT + 15 S/R

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Vidush IIITD
19 September, 2:47 pm



between the transmission of x and the transmission of x+1, then it will be a long time until x can be recovered, under a NAK only protocol.

- The crucial point is long delay before recovery. Rest of the details may change but student must point out the long delay.

On the other hand, if data is being sent often, then recovery under a NAK-only scheme could happen quickly. Moreover, if errors are infrequent, then NAKs are only occasionally sent (when needed), and ACK are never sent – a significant reduction in feedback in the NAK-only case over the ACK-only case.

- The crucial point is quick delivery.

4. Consider transferring an enormous file of L bytes from Host A to Host B. Assume an MSS of 536 bytes.
   a. What is the maximum value of L such that TCP sequence numbers are not exhausted? Recall that the TCP sequence number field has 4 bytes. (2 marks)
   b. For the L you obtain in (a), find how long it takes to transmit the file. Assume that a total of 66 bytes of transport, network, and data-link header are added to each segment before the resulting packet is sent out over a 155 Mbps link. Ignore flow control and congestion control so A can pump out the segments back to back and continuously. [2 marks]

There are $2^{32} = 4{,}294{,}967{,}296$ possible sequence numbers.

a) The sequence number does not increment by one with each segment. Rather, it increments by the number of bytes of data sent. So the size of the MSS is irrelevant -- the maximum size file that can be sent from A to B is simply the number of bytes representable by $2^{32} \approx 4.19\,\text{Gbytes}$.

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